FederAI Cloud — Reviewer Demo & Technical Guide

A Cloud-Native, Secure and Privacy-Preserving Infrastructure for Federated AI

---

1. Executive Summary & Problem Solved

Core Challenge: Traditional centralized Machine Learning requires aggregating patient health records into a central server. In healthcare and finance, this violates regulatory privacy mandates (HIPAA, GDPR) and introduces massive data leak liabilities.

The FederAI Cloud Solution: *'Train Together. Share Intelligence, Not Data.'*
The model architecture travels to edge hospital nodes. Hospitals compute local weight gradients on private data. Gradients pass through Differential Privacy (DP-SGD) noise injection, Directional Cosine alignment verification, and Statistical L2-Norm Z-score filtering before being aggregated into a Byzantine-resilient global clinical risk model.

2. Step-by-Step Reviewer Demonstration Script

Step 1: Open the Control Plane

Navigate browser to **http://localhost** (Port 80). Show the live glassmorphism HUD, Active Clients (5), Global Accuracy (50.0%), and nominal status. Point out the zero-data-leakage architecture.

Step 2: Inspect 5 Hospital Nodes

Click **Federation Map & Nodes** tab. Point out 5 medical centers (Mount Sinai NY, Mayo Clinic, Stanford, Johns Hopkins, Cleveland Clinic) with independent non-IID clinical risk datasets and initial trust scores.

Step 3: Run Honest Federated Training

Go to **Training Controls**. Set Target Rounds = 5, Strategy = FedAvg, DP = Enabled (ε=1.0, δ=10■■). Click **Launch Federation Round**. Watch the WebSocket state machine transition through Dispatch → Train → DP → Threat Scan → Aggregate. Show validation accuracy climbing to ~80-85%.

Step 4: Byzantine Threat Chaos Injection

Under **Training Controls**, toggle *Adversarial Attack Simulation* to **ON**. Select **Sign-Flipping Gradient Inversion** or **Gradient Scaling (20x)**. Click **Launch Federation Round**.

Step 5: Show Live Byzantine Defense

Show Hospital Delta turning **BRIGHT RED** in quarantine. Switch to **Security Center** tab: show the Directional Cosine Anomaly check (< 0.0) or Norm Z-Score (> 3.0), and the Bayesian Trust score penalty. Emphasize that global model accuracy was protected because the poisoned gradient was discarded!

Step 6: Zero-Downtime Model Rollback

Go to **Model Registry** tab. Show immutable S3/SeaweedFS model checkpoints (v1, v2). Click **Rollback** on model-clinical-v1 to demonstrate instant production rollbacks.

Step 7: Cloud Telemetry & Cost Modeling

Go to **Infrastructure** and **Cost Estimator** tabs to review live Prometheus metrics, Grafana observability, and AWS monthly cost breakdowns across EKS, RDS Multi-AZ, and S3.

3. Technologies & Tools Used in FederAI Cloud

Below is the complete engineering matrix of every technology used and its specific architectural purpose:

Component / Tool	Category	Exact Architectural Role in FederAI Cloud
Python 3.11 & NumPy	Machine Learning	Core mathematical engine: ClinicalRiskMLP neural network forward/backward pass, SGD optimizer, gradient clipping.
Differential Privacy (DP-SGD)	Cybersecurity / AI	Gaussian & Laplace noise perturbation, L2 gradient clipping (C=1.0), Moments Accountant tracking.
FastAPI & Uvicorn	Backend Gateway	Asynchronous REST API Gateway, WebSocket live event dispatcher, JWT auth, RBAC permissions, /metrics endpoint.
PostgreSQL 16 & SQLAlchemy	Relational Database	Multi-tenant persistence for organizations, client nodes, sessions, training rounds, audit logs, and security events.
SeaweedFS / AWS S3	Object Storage	S3-compatible object storage for immutable binary model checkpoints (v1, v2), gradient tensors, and instant rollbacks.
RabbitMQ 3.12	Message Broker	AMQP async event broker for coordinating distributed round dispatch and aggregation worker synchronization.
Prometheus & Grafana	Observability	Time-series metrics scraping from FastAPI and real-time visualization dashboards for ML convergence and server health.
Docker & Compose	DevOps Containerization	Multi-stage container builds and single-command local cloud orchestration with automated healthchecks.
Kubernetes (EKS) & HPA	Cloud Orchestration	Declarative manifests, Horizontal Pod Autoscalers (70% CPU/RAM), Ingress routing, StatefulSets, and PVC storage.
Terraform (AWS)	Infrastructure as Code	Modular AWS cloud infrastructure: Multi-AZ VPC, EKS Managed Node Groups, RDS Multi-AZ PostgreSQL, KMS S3, ALB.
HTML5 / CSS3 / ES Modules	Frontend SPA	Custom dark glassmorphism Command Center SPA with zero heavyweight frameworks, live WebSockets, and chart canvas.

4. Comprehensive Guide to Every Button & Input in the UI

This reference explains the precise functionality of every interactive element across all screens:

UI Location	Button / Input Control	Exact Functional Action & Backend Effect
Sidebar	**Command Center Tab**	Switches to the main executive overview HUD, active session status, convergence trends, and quick action cards.
Sidebar	**Federation Map Tab**	Displays the 5 hospital geographic nodes, individual client status badges, dataset sizes, and Bayesian trust scores.
Sidebar	**Training Controls Tab**	Opens round orchestrator configuration, strategy selection, Differential Privacy parameters, and attack injectors.
Sidebar	**Model Registry Tab**	Lists all immutable global model versions in S3 storage with accuracy scores, validation loss, and rollback buttons.
Sidebar	**Security Center Tab**	Displays Byzantine threat detection logs, directional cosine similarity scores, norm Z-scores, and quarantine events.
Sidebar	**Infrastructure Tab**	Shows real-time CPU, RAM, Network I/O, database latency, and replica counts for all 7 platform microservices.
Sidebar	**Cost Estimator Tab**	Provides dynamic AWS infrastructure cost calculators based on node cluster sizes and training frequency.
Sidebar	**Viva & Reviewer FAQ**	Opens the interactive academic examiner question-and-answer guide with formal mathematical definitions.
Top Header	**Start Training Button**	Sends POST /api/federation/start to trigger the next federated training round across all active clients.
Top Header	**Reset Session Button**	Sends POST /api/federation/reset to reset training session back to Round 0 and baseline model v1.
Top Header	**Telemetry Badges**	Live WebSocket indicators showing active round count, global accuracy, convergence loss, and privacy budget spent.
Training Controls	**Target Rounds Input**	Sets maximum number of iterative federated rounds to execute (default: 20 rounds).
Training Controls	**Target Accuracy Input**	Defines target validation accuracy threshold (e.g. 0.88) at which training will automatically complete.
Training Controls	**Strategy Dropdown**	Selects aggregation algorithm: **FedAvg** (sample-weighted), **Trimmed Mean** (coordinate-trimmed), or **Multi-Krum** (distance-based).
Training Controls	**DP Epsilon (ε) Input**	Sets Differential Privacy budget parameter (e.g. 1.0). Smaller values inject more noise for stronger privacy.
Training Controls	**Attack Toggle Switch**	Enables simulation of Byzantine adversarial behaviors from Hospital Delta during the upcoming round.
Training Controls	**Attack Vector Dropdown**	Selects specific threat mode: **Sign-Flipping**, **Gradient Scaling (20x)**, **Random Noise**, or **Label Flipping**.
Training Controls	**Launch Round Button**	Executes a discrete training round synchronously, displaying live step tracker progress (1 through 5).

Federation Map	**Node Cards (Hospital Alpha-Epsilon)**	Clicking any hospital card opens the deep-dive node inspector showing IP address, EHR cohort size, and trust history.
Federation Map	**Unquarantine Button**	Sends POST /api/clients/{id}/unquarantine to restore a quarantined node after manual security review.
Model Registry	**Rollback Button**	Sends POST /api/models/rollback to restore active global model weights to the selected previous version in S3.
Model Registry	**Download Checkpoint**	Retrieves pre-signed binary S3 download URL for local model inspection or offline inference testing.
Infrastructure	**Simulate Node Failure**	Sends test chaos event to verify Kubernetes pod self-healing and HPA replica scaling triggers.